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Contextual Pulls for Synchrony and Segregation: An Empirical Test of a Novel Theoretical Framework

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Interpersonal synchrony—the temporally coordinated alignment of behavior, physiology, and neural activity—is crucial for successful social interaction and group performance. However, the context-dependent nature of synchrony remains underexplored. This study offers an empirical test of the multimodal flexible synchrony theory, which posits that social contexts are defined by two latent potentials: the pull toward synchrony and the pull toward segregation. According to the theory, these latent pulls shape the expression and function of multimodal synchrony. To test this theory experimentally, these pulls were manipulated in a dyadic coordination task, and their effects on the relationship between two indices of physiological synchrony, performance, and group cohesion were examined. Results revealed that the relationship between physiological synchrony and performance and cohesion depends on the balance of these contextual pulls. In high-segregation contexts, heart rate interbeat interval (IBI) synchrony was negatively associated with performance, while in low-segregation contexts, IBI synchrony was positively associated with performance. These results were also true for the association between IBI synchrony and cohesion but only in high-synchrony contexts. Electrodermal activity (EDA) synchrony was positively associated with performance in high-segregation or low-synchrony contexts and was negatively associated with performance in low-segregation or high-synchrony contexts. EDA synchrony was not associated with group cohesion. These findings provide validation for the flexible multimodal theory and have significant implications for understanding and optimizing human collaboration across various social settings, suggesting the importance of considering both the need for alignment and individual autonomy.

Public Significance Statement

This research highlights the critical role of context in shaping human interaction, emphasizing the need to balance shared action with individual autonomy. Understanding this dynamic interplay provides valuable insights for improving teamwork, communication, and overall performance in various settings, from workplaces to educational environments.

Keywords: interpersonal synchrony, multimodality, cohesion, performance, physiological synchrony

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All data are publicly available on the Open Science Framework at <https://osf.io/349su/>, and the experimental task is available on GitHub at <https://github.com/OdedMayo/The-Herding-Task>.

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Human and animal survival depends on coordinating actions effectively (Freeberg et al., 2019; Keeble et al., 2022). From schools of fish and flocks of birds to human groups gathering food or playing team sports, coordination enables adaptation to complex and dynamic environments. However, effective coordination is not a one-size-fits-all process; individuals, dyads, and groups must adapt to the specific demands of diverse contexts (Kameda & Tindale, 2013). At times, this adaptation requires tight coordination across multiple levels—actions, cognitive processes, emotions, or even physiological states. For example, hunting together requires synchronized actions, while problem solving in a team might necessitate shared cognitive processes or emotional alignment. Conversely, certain contexts demand individual autonomy, such as dividing labor during a group task or generating unique ideas during a brainstorming session.

Interpersonal Synchrony and Contextual Dynamics

Interpersonal synchrony is the temporal rhythmic coordination in behavior, physiology, or cognition (Ackerman & Bargh, 2010; Bernieri & Rosenthal, 1991; Palumbo et al., 2017). In this field of research, the context's critical role has become increasingly recognized (Danyluck & Page-Gould, 2019; Gordon et al., 2025; Mayo & Gordon, 2020; Mayo et al., 2023), influencing the level of synchrony as well as its adaptiveness. For example, research shows that cooperative versus competitive contexts alter the meaning and outcomes of physiological synchrony (Danyluck & Page-Gould, 2019), while affiliative conversations elicit more behavioral synchrony than argumentative ones (Paxton & Dale, 2013). These findings highlight the importance of social context, yet current characterizations of context often rely on specific attributes, such as emotional tone, affiliative intent, or goal orientation.

Although these attributes provide valuable insights, they do not address the fundamental question: *What underlying principles govern the influence of context on interpersonal synchrony and its meaning?* To answer this question, recent theoretical works (Gordon et al., 2025; Mayo & Gordon, 2020) suggest that most social contexts can be understood through two latent potentials: the pull toward synchrony and the pull toward segregation. A central theoretical claim of the multimodal flexible synchrony framework (Gordon et al., 2025; Mayo & Gordon, 2020) is that the contextual pulls for synchrony and segregation are not opposite ends of a single continuum but rather two distinct latent dimensions that can co-occur in varying degrees. For example, in friendly competition or collaborative problem solving, individuals may need to synchronize around shared goals while also expressing autonomous contributions. Such contexts are termed *metastable* (see also Tognoli et al., 2018) as they move between periods of quasistable synchrony and periods of asynchronized activity. Metastable contexts are everywhere. Consider a brainstorming session: Participants must generate independent ideas (segregation) while simultaneously building mutual understanding through shared discussions (synchrony).

Investigating Latent Contextual Features

This study aimed to experimentally test, for the first time, how experimental manipulation of these two latent potentials—pull to synchrony and pull to segregation—affects interpersonal synchrony and its associations with performance and social outcomes. By moving beyond specific situational attributes, this research seeks to uncover the fundamental contextual features that shape the meaning and adaptive function of interpersonal synchrony.

This focus on latent potentials offers a promising conceptual lens for investigating how social context shapes synchrony, extending beyond traditional task-based or situational accounts. Existing theories often emphasize the situational specifics of synchrony, such as affiliative intent or competitive dynamics, but this framework highlights the underlying mechanisms that drive adaptation across diverse settings. By proposing synchrony and segregation pulls as fundamental features of context, this work offers an initial step toward advancing our understanding of how social interaction operates across modalities and levels of analysis. By identifying the latent features of context that shape synchrony's meaning, this study may offer insights with potential relevance to real-world coordination challenges.

Distinct Outcome Measures

Recently, it has been acknowledged that when investigating the outcomes of interpersonal synchrony, one should take into account various types of outcomes (Galbusera et al., 2019; Gordon et al., 2025). There are two main group outcomes: the



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social/relationship outcomes and the performance outcomes. At times, these two types of outcomes are in line, while at other times, they diverge (Mayo et al., 2021). For example, in a classroom, synchrony could likely enhance both the emotional atmosphere and learning processes. On the other hand, during brainstorming sessions, high synchrony might feel good but might also be associated with groupthink.

Experimental Design and Contextual Manipulation

In this study, the pull toward synchrony refers to the contextual potential for individuals to align their actions. Conversely, the pull toward segregation reflects the contextual potential for individuals to act independently or divergently, emphasizing autonomy or differentiation. Note that the potential of the context means the extent to which the context pulls the individuals in it to certain states, in this case, of synchrony or segregation. By systematically manipulating these pulls, we created experimental conditions that allowed us to observe how these latent potentials influence levels of behavioral and physiological synchrony, as well as their association with performance and cohesion.

To explore these dynamics, we adapted an established computerized coordination task (Nalepka et al., 2017), in which participants work together to “herd” sheep in a digital interface. Prior research has shown that this task naturally elicits coordination, even without explicit communication. Our contribution lies in modifying this task to experimentally manipulate contextual demands—specifically, the pull toward synchrony and the pull toward segregation—creating a 2×2 design with four distinct contexts (high/low synchrony and high/low segregation). This approach enabled us to isolate the influence of these latent contextual features on multimodal synchrony and its downstream effects on task outcomes.

Multimodal Approach

Synchrony can occur across multiple modalities, including behavioral synchrony, physiological synchrony, and inter-brain synchrony (Ackerman & Bargh, 2010; Gordon et al., 2025; Mayo & Gordon, 2020). Each of these modalities may serve distinct functions in various contexts (Mayo et al., 2021; Reindl et al., 2022), although certain outcomes, such as social cohesion, may arise through multiple pathways (Gordon et al., 2025). For example, neural synchrony might enhance shared understanding during a team task (Wheatley et al., 2012), while physiological synchrony could reflect emotional attunement in therapeutic settings (Atzil-Slonim et al., 2023). Understanding these distinct yet interconnected modalities is critical for comprehending how synchrony operates and adapts to contextual demands.

In the present study, we focused on two widely used autonomic indices of physiological synchrony (see Palumbo et al., 2017): interbeat interval (IBI) synchrony and electrodermal activity (EDA) synchrony. Although both measures are influenced by arousal and can co-occur, they differ meaningfully in neural innervation, temporal dynamics, and psychophysiological function. IBI reflects the time between heartbeats and is influenced by both the parasympathetic and sympathetic branches of the autonomic nervous system (Berntson et al., 1997). In many contexts, IBI synchrony is interpreted as a marker of coregulation, affective bonding, or sustained interpersonal engagement (Feldman, 2012; Palumbo et al., 2017). However, it has also been observed during co-activation and arousal states, such as shared laughter or intense ritual performance (Helm et al., 2014; Konvalinka et al., 2011; Lackner et al., 2019; Palumbo et al., 2017), suggesting that it reflects broader forms of internal alignment. Notably, IBI synchrony can also emerge in contexts with a high pull for segregation, such as friendly competition (Danyluck & Page-Gould, 2019) or parallel individual action (Fusaroli et al., 2016), where individuals may still seek a shared emotional or cognitive connection. This underscores the importance of considering context when interpreting the function of synchrony.

By contrast, EDA is driven solely by sympathetic nervous system activity and reflects changes in skin conductance that can be decomposed into two components: a slow-changing tonic signal (skin conductance level, SCL), which indexes general arousal, and a faster, phasic signal (skin conductance response, SCR), which reflects transient reactions to emotionally or cognitively salient events (Critchley, 2002; Dawson et al., 2007). Because the current paradigm involved continuous, nonevent-related social interaction, we focused on overall covariation in the tonic EDA signal, which captures shared fluctuations in general sympathetic arousal over time. This approach captures shared physiological engagement and has been used in prior interpersonal synchrony research (e.g., Gordon et al., 2023). EDA synchrony has been linked to

shared attentional focus, arousal, or stress, depending on the context (Henning et al., 2001; Mønster et al., 2016), and tends to reflect moment-to-moment alignment in physiological engagement rather than discrete event-related peaks. It is more reactive and event linked than IBI, making it particularly sensitive to dynamic changes in engagement during cognitively demanding tasks.

These physiological and temporal differences suggest that IBI and EDA synchrony capture complementary modes of interpersonal alignment. We propose that their relationship to performance and cohesion is shaped by the latent contextual pulls for synchrony and segregation. Specifically, we hypothesize that IBI synchrony will be positively associated with task performance and social cohesion in contexts with a high pull for synchrony or a low pull for segregation, where explicit coordination is beneficial. However, in contexts with a low pull for synchrony or a high pull for segregation, we hypothesize that IBI synchrony will be negatively associated with outcomes, as excessive alignment may impair task-appropriate independence or flexibility.

Conversely, we hypothesize that EDA synchrony will be positively associated with performance in contexts that require autonomy (high segregation) or do not promote overt coordination (low synchrony), where shared physiological arousal may compensate for the lack of behavioral synchrony. In contexts with a high pull for synchrony or low pull for segregation, however, we expect EDA synchrony to be unrelated or even negatively related to performance, possibly reflecting overstimulation or redundant arousal. Given the mixed findings in the literature regarding EDA synchrony and social outcomes (e.g., Behrens et al., 2020; Chikersal et al., 2017; Gordon, Wallot, & Berson, 2021), we explore its association with cohesion without a priori directional predictions.

Finally, based on prior research, we wanted to examine the association between IBI and EDA synchrony under varying contexts (pulls for synchrony and segregation). Prior research has shown that the underlying physiological signals that drive EDA and IBI are often interrelated. For example, in a study that examines stress responses, an increase in the SCR frequency and a decrease in high-frequency components of heart rate variability were found (Yu et al., 2024). Another study examined the association between heart rate and EDA during the Rorschach test and found that these activities were indeed synchronized (Kettunen et al., 1998). Golland et al. (2014) examined physiological responses to emotional movies. They also found an association between EDA and heart rate activity. These findings highlight the importance of considering both physiological synchrony and the potential covariation or divergence between EDA and IBI, particularly in multimodal analyses. Because the influence of the context is yet unknown, this analysis will be exploratory.

Method

Participants

The sample size was determined through an a priori power analysis using G*Power 3.1 (Faul et al., 2009). We anticipated a small effect size ($d = 0.15$), which represents a conservative estimate for studies of social interaction (Ferguson, 2016). Although our main analyses employed multilevel models (MLMs) with dyads and repeated measures, formal power analysis tools for such designs are limited and require assumptions about variance components and intraclass correlations that are difficult to estimate reliably in novel paradigms (Murayama et al., 2022).

As an approximation, we used a paired-sample (within-subject) t test in G*Power with $d = 0.15$, $\alpha = .05$ (two-tailed), and 80% power, which indicated a required sample size of 351 participants. Given that the present study used a multilevel design with repeated observations per participant and per dyad, this analysis should be viewed as a conservative upper bound. We therefore aimed to recruit as many participants as feasible within the resource constraints of the project and ultimately included 92 participants (46 dyads). Notably, while sample sizes vary across studies investigating physiological synchrony, several prior investigations have used samples in the 40–60 dyad range (e.g., Bar-Kalifa et al., 2019; Cook, 2020; Creavy et al., 2020; Li et al., 2021; Lin et al., 2024). Accordingly, our sample of 46 dyads is comparable to other works in this domain.

While our synchrony predictors (e.g., IBI, EDA) were computed at the dyad level, key outcome variables (e.g., cohesion) were measured at the participant level. All analyses were conducted at the participant level, with dyad modeled as a random effect to account for nonindependence. This approach provided a consistent analytic framework across outcomes. The robust and theoretically meaningful effects observed across models further support the adequacy of our sample size.

Ninety-two participants (19 males, 21%; 73 females, 79%) were randomly assigned to 46 same-gender dyads. All participants were undergraduate psychology students from the Department of Psychology at Bar-Ilan University and received course credit in exchange for their participation. Inclusion criteria required participants to be between the ages of 18 and 40, have no known heart conditions (which could influence IBI measurement), and refrain from consuming caffeine or engaging in aerobic activity in the hours prior to the lab session. The mean age was 22.9 years ($SD = 2.18$, range = 19–33).

Dyads were formed during a prescreening phase. Only same-gender pairs were selected for participation, and when there was an odd number of potential participants of a given gender, one participant from that group was not invited to the lab. As a result, all dyads were composed of participants who matched in gender and were randomly paired within gender groups.

Regarding background characteristics, 42 participants had a secondary education level (45.65%), 14 had postsecondary

education (15.21%), and 32 had undergraduate education (34.78%). Forty-five participants reported a low-income level (48.91%), nine reported a moderate-income level (9.78%), and 37 were currently unemployed (40.21%). In terms of religiosity, 31 participants identified as secular (33.70%); 17 as culturally, but not strictly, religious (18.48%); 40 as religious (43.48%); and four as other (4.34%).

Informed consent was obtained before participation. Participants also completed online questionnaires at home before the lab visit; however, these data are not included in the present study.

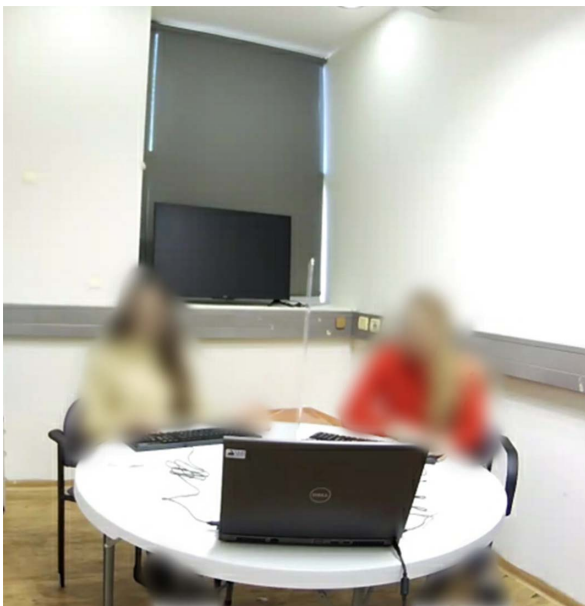
Procedure

The study was approved by the Bar-Ilan University Department of Psychology Ethics Committee (No. 2020/40) and followed ethical guidelines. Participants were instructed to stay hydrated and refrain from consuming caffeine or nicotine for at least 2 hr before arrival. Upon arrival, participants were informed that they would be connected to physiological monitoring equipment and recorded on video for data analysis.

A research assistant (RA) provided an overview of the study and attached physiological sensors to each participant using MindWare Mobile Recorders (MindWare Technology, Gahanna, OH). Participants were then seated at a table in front of a shared computer screen, each with a keyboard, and separated by a transparent glass divider due to COVID-19 regulations (see Figure 1). The RA provided standardized instructions:

Figure 1

The Setting of the Herding Task



Note. See the online article for the color version of this figure.

Welcome. During the next half hour, you will perform a dyadic task using the keyboard. Please refrain from speaking to each other during the experiment.

After a 5-min baseline period to verify data recording, participants completed the herding task, followed by a questionnaire assessing perceived social cohesion. Finally, participants were debriefed.

The Herding Task

The herding task is a virtual dyadic game adapted from the biagent sheep herding game (Nalepka et al., 2017) that was designed using the Unity 3D game engine (Version 2018.2beta; Unity Technologies, San Francisco, CA). Each dyad worked together to guide virtual sheep using their respective sheepdogs to keep the sheep within a designated circular area on the screen. The sheep moved randomly but reacted to the sheepdogs' proximity, moving away at a speed proportional to the distance from the sheepdogs. If a sheep left the circular area, it was reset to the center, and the trial restarted.

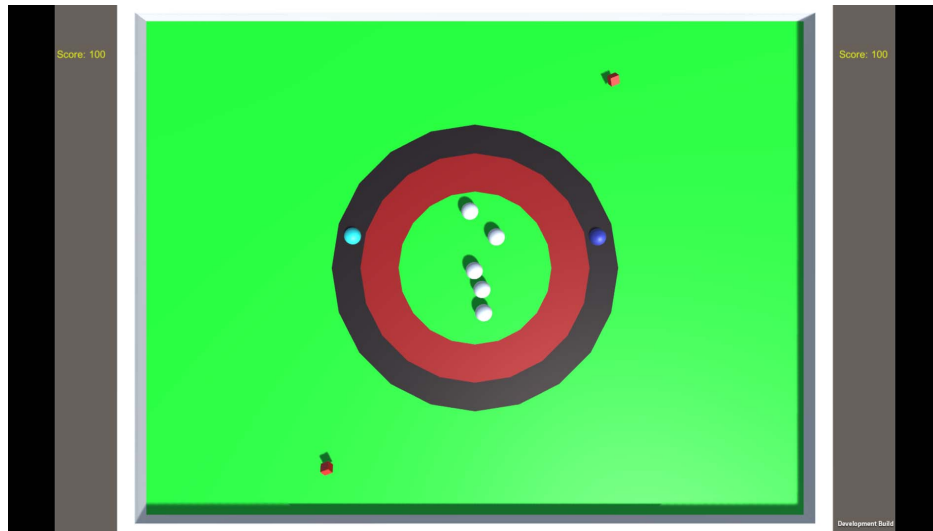
Research has shown that dyads tend to develop spontaneous synchronized herding behaviors, even without verbal communication (Nalepka et al., 2017). However, our study modified the game to incorporate an additional segregation component, requiring participants to independently collect on-screen targets while also managing the sheep. Two randomly placed targets appeared at a time, one per player's side of the screen. Picking up a target caused it to reappear elsewhere on the same side. Figure 2 illustrates the herding task.

Unlike Nalepka et al. (2017), where coordination strategies emerged spontaneously, our paradigm was designed to bias dyads toward specific coordination demands through experimental manipulation. Participants were required to learn a shared containment strategy during training, ensuring that all dyads entered the experimental conditions with a baseline coordination ability. The independent variables (sheep speed and target frequency) then modulated the pull toward synchrony and segregation, allowing us to test their interactive effects on behavioral and physiological alignment.

To manipulate the relative pulls for synchrony versus segregation, we varied (a) sheep speed (higher speeds requiring greater coordination) and (b) target frequency (higher frequencies increasing independent actions; high frequency was every .22 Hz, and low frequency was .02 Hz). This 2×2 within-subject design allowed us to systematically examine how varying levels of synchrony and segregation demands influenced dyadic coordination.

Each participant controlled their sheepdog using a keyboard: The up/down keys moved the sheepdog along a circular perimeter surrounding the sheep's central area; the left/right keys were used to collect targets. When pressed, the sheepdog moved directly toward the target and then returned upon a second key press. Each task phase was preceded by instructions

Figure 2
A Snapshot of the Screen at the Start of the Game



Note. The dark blue and light blue balls are two “sheepdogs.” The white balls at the center of the screen are the sheep. The red circle and the green circle inside it are the circular areas of the sheep. The black circle around them is the track upon which the “sheepdogs” move. The red cubes are the targets. The objective of the task is for participants to coordinate their movements to keep the sheep inside the goal area while simultaneously collecting targets with their sheepdogs. See the online article for the color version of this figure.

from the RA, who ensured participants understood the procedure before starting.

Training Phases

Before the experimental trials, participants completed three training phases:

1. Training Phase 1 (sheep only): Dyads controlled sheepdogs without targets. They had to keep all the sheep in the circular area for 30 s. If a sheep left, the trial restarted.
2. Training Phase 2 (targets only): Dyads collected six targets without sheep.
3. Training Phase 3 (full task): Dyads practiced with both sheep and targets for 2 min. The RA explained the scoring system: +1 point per collected target, −3 points per escaped sheep, and the total score that is displayed in real time. This scoring system aims to provide a balance between the synchronization and segregation tendencies.

Experimental Conditions

Dyads completed four experimental conditions (2 min each), presented in random order:

- low sheep speed, low target frequency (low pull for synchrony, low pull for segregation)
- low sheep speed, high target frequency (low pull for synchrony, high pull for segregation)
- high sheep speed, low target frequency (high pull for synchrony, low pull for segregation)
- high sheep speed, high target frequency (high pull for synchrony, high pull for segregation)

Physiological Measures

Electrocardiogram Collection and Preprocessing

Heart activity was recorded using a modified lead II electrocardiogram configuration (Sherwood et al., 1990). Electrodes were connected to MindWare Mobile Recorders, which wirelessly transmitted data to a control-room laptop. Data were sampled at 500 Hz. Electrocardiogram signals were processed using MindWare heart rate variability Analysis Software (v3.1.4). Graduate students visually inspected and manually corrected artifacts (Berntson et al., 1997). Signals were amplified (gain = 1,000) and filtered using a Hamming windowing function. The IBI time series was spline interpolated at 500-ms intervals using MATLAB (MathWorks, 2023).

EDA Collection and Preprocessing

EDA was collected at a sampling rate of 500 Hz using two Ag/AgCl electrodes placed on the nondominant palm of each participant. Signals were processed using MindWare Technologies' EDA application software (Version 3.1.5) and were visually inspected and manually edited by trained researchers to remove artifacts. The signal was smoothed using a rolling filter of 500 data points per block. We focused on the overall SCL, which integrates both tonic and phasic (SCR) components of EDA activity. Because our design was not event related and we had no a priori hypotheses about differential mechanisms underlying the tonic and phasic components, we did not detrend or separate these signals. This approach follows prior studies of EDA synchrony during continuous social interaction, in which the goal is to capture overall covariation in sympathetic arousal rather than event-locked responses (e.g., Gordon et al., 2023). SCL was downsampled to 2 Hz by averaging values in 500-ms bins, and values were recorded in microsiemens (μS), with a minimum threshold of .05 μS to exclude signal floor noise. This threshold corresponds to the default setting in MindWare and has been used in prior work (e.g., Gordon, Horeish, et al., 2021).

IBI Synchrony Calculation

IBI synchrony was computed as the Pearson correlation between dyad members' interpolated IBI time series within each condition. Because we were interested in the strength of covariation in autonomic regulation rather than the specific direction (i.e., in phase vs. antiphase), we used the absolute value of the correlation coefficient. This approach reflects the extent to which participants' arousal or regulation levels rise and fall together over time, regardless of whether those changes occur in synchrony or mirrored patterns. Such patterns are both indicative of coordinated physiological dynamics.

This method has been used to assess dyadic physiological synchrony in interactive contexts (for review, see Mayo et al., 2021) and has been shown to capture meaningful interpersonal coordination across a variety of emotional and behavioral settings.

EDA Synchrony Calculation

To compute EDA synchrony, we applied the same method as for IBI: We calculated the Pearson correlation between the downsampled EDA time series of each dyad member within each condition. As with IBI, we used the absolute value of the correlation to capture the strength of covariation in autonomic arousal, regardless of its direction, a method that has been previously used in EDA synchrony analysis (Gordon et al., 2023; Mayo et al., 2021). In this study, the computed synchrony therefore reflects covariation in the overall SCL signal (integrating both tonic and phasic changes) rather than in discrete SCRs.

Behavioral Measures

Performance Score

Each dyad began each 2-min condition with a score of 100 points. During the task, the score was dynamically adjusted in real time: Participants earned +1 point per collected target and lost -3 points per sheep escape. This scoring system was designed to reflect performance trade-offs between joint coordination (sheep containment) and independent action (target collection). The total score at the end of each condition was used as the main performance measure. To account for differences in baseline difficulty across the four conditions and because there were considerable differences between scores within the conditions, we centered the scores around the mean within each condition.

Behavioral Synchrony

Behavioral synchrony was assessed by tracking the position of each player's sheepdog on the screen at a sampling rate of 2 Hz. Synchrony was defined as the duration of time in which both players were located on the circular perimeter surrounding the sheep area simultaneously for at least 2 consecutive seconds. This threshold was chosen to ensure that behavioral coordination reflected intentional joint containment behavior rather than brief or incidental overlap. The synchronized perimeter movement corresponds to the containment strategy practiced during the training phase and is essential for preventing sheep escape. Periods not classified as behavioral synchrony typically reflected task-driven divergence, namely, when one or both players moved toward targets. While movement synchrony could also be conceptualized in more continuous terms (e.g., directional alignment, speed matching, or synchronized keypresses), we faced a conceptual challenge in combining these continuous features with the binary spatial constraint (either both players were on the perimeter or not). To preserve conceptual clarity and ensure alignment with the core task demands, we adopted this categorical operationalization.

Surveys

Demographics

Before the lab visit, participants completed an online Qualtrics survey, reporting their age, gender, education level, employment status, and medical history (cardiac conditions, medication use).

Group Cohesion

After the task, participants completed an eight-item cohesion scale (Gordon et al., 2020; Podsakoff & MacKenzie, 1994) assessing teamwork (e.g., "We worked together as a team") on a 1-6 Likert scale. Scores were treated as individual-level

responses rather than being averaged across dyad members, as cohesion reflects a subjective interpersonal experience. The scale's reliability was high (Cronbach's $\alpha = .89$; McDonald's $\omega = .89$).

Missing Data

The data of one dyad include only measures from three (out of four) conditions due to technical difficulties in data recording. In addition, five individual dyad-condition EDA recordings were excluded due to technical problems: in three of these, the EDA device failed to record; in two others, the recording was too short (more than 5 s of the 120-s condition were missing), falling below our reliability threshold.

Statistical Analysis

We used linear mixed-effects models (MLM) to test our hypotheses, conducting analyses in jamovi (Version 2.4.11). First, we tested the effectiveness of our manipulation, examining whether the pull for synchrony, the pull for segregation, and their interaction influenced behavioral synchrony. Next, we assessed the effects of IBI synchrony on task performance, testing whether this relationship was moderated by the pull for synchrony and segregation (and their interaction). We then examined whether IBI synchrony was associated with cohesion and whether this effect was similarly moderated.

Parallel analyses were conducted for EDA synchrony, assessing its associations with performance and cohesion under different contextual conditions. Finally, we examined the relationship between IBI synchrony and EDA synchrony, treating EDA synchrony as the dependent variable, and tested whether this association was moderated by the pull for synchrony and segregation (and their interaction).

To account for the dyadic structure of the data, we included random intercepts and slopes for condition order, the pull for synchrony, and the pull for segregation at the dyad level. Interaction effects were analyzed by plotting the estimated slopes at each level of the pull for synchrony and segregation (low, high). For three-way interactions, we visualized the slopes across all combinations of these contextual factors. In all our analyses, all the independent variables were centered to their mean. To control for multiple comparisons, we applied false discovery rate correction (Benjamini & Hochberg, 1995) within each model, adjusting for all effects of interest.

Transparency and Openness

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. All data are openly available on the Open Science Framework

(Mayo & Gordon, 2025a), and the experimental task is available on GitHub (Mayo & Gordon, 2025b). Data were analyzed using R (Version 4.4.1), jamovi (Version 2.4.11), and MATLAB (Version R2023b). This study's design and its analysis were not preregistered.

Results

Manipulation Check

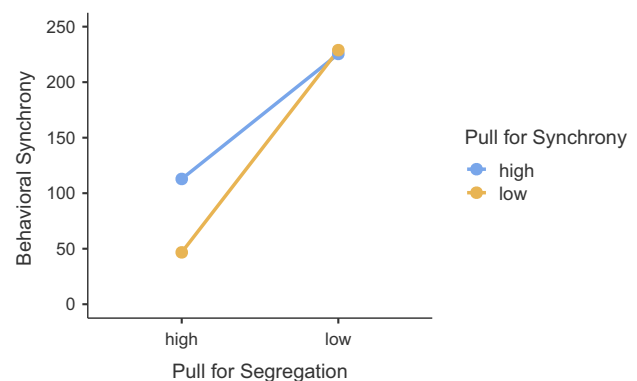
To verify our manipulation, we examined whether the pulls for synchrony and segregation were indeed associated with the level of behavioral synchrony. An MLM model was used (Equation 1):

$$\begin{aligned} \text{Behavioral Synchrony} \sim & 1 + \text{Pull for Synchrony} \\ & + \text{Pull for Segregation} + \text{Order} \\ & + \text{Pull for Synchrony} \\ & \times \text{Pull for Segregation} \\ & + (1 + \text{Pull for Synchrony} \\ & + \text{Pull for Segregation} \\ & + \text{Order} | \text{Dyad}). \end{aligned} \quad (1)$$

Results indicate that our model accounted for 76.2% of the variance of behavioral synchrony (see Supplemental Table S1 and Figure S1). The pulls for synchrony and segregation explained the vast majority of the variance in behavioral synchrony in the task. As hypothesized, there was a main effect for the pull for synchrony ($F = 17.09, p < .001$) and a main effect for the pull toward segregation ($F = 544.52, p < .001$). The order of the conditions did not have an effect of behavioral synchrony ($F = .36, p = .55$). As shown in Figure 3, behavioral

Figure 3

Behavioral Synchrony (in Half Seconds) as a Function of the Pull for Segregation and the Pull for Synchrony



Note. The y-axis represents the level of behavioral synchrony, while the x-axis represents conditions varying in segregation (high vs. low). The different lines indicate conditions of high and low pull for synchrony. As expected, behavioral synchrony increases with higher pull for synchrony and decreases with higher pull for segregation. See the online article for the color version of this figure.

synchrony increased under high synchrony pulls and decreased under high-segregation pulls. Additionally, an ordinal interaction effect Pull for Synchrony $\times$ Pull for Segregation ($F = 1182.69, p < .001$) indicates that the difference in behavioral synchrony between high- and low-synchrony conditions was greater when the pull for segregation was also high. That is, the effect of the pull for synchrony was amplified in high-segregation contexts. Notably, the effects were additive and interacting in a theoretically consistent manner.

The Associations Between Task Performance and IBI Synchrony Depend on the Context's Requirement to Segregate

We first examined how the association between IBI synchrony and task performance varied depending on the context's pull for synchrony and segregation. An MLM model was used (Equation 2):

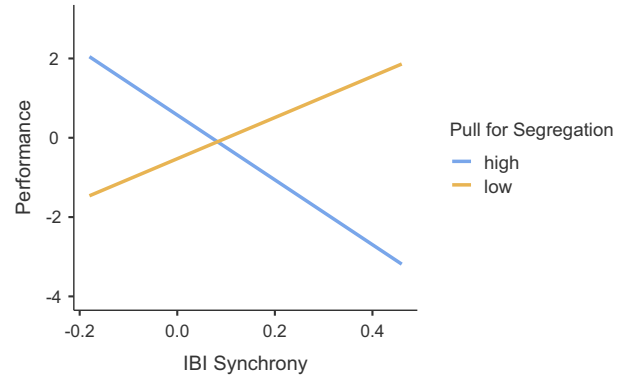
$$\begin{aligned} \text{Task Performance} \sim & 1 + \text{Pull for Synchrony} \\ & + \text{Pull for Segregation} + \text{Order} \\ & + \text{IBI Synchrony} \\ & + \text{Pull for Synchrony} \\ & \times \text{Pull for Segregation} \\ & + \text{IBI Synchrony} \\ & \times \text{Pull for Synchrony} \\ & + \text{IBI Synchrony} \\ & \times \text{Pull for Segregation} \\ & + \text{IBI Synchrony} \\ & \times \text{Pull for Synchrony} \\ & \times \text{Pull for Segregation} \\ & + (1 + \text{Pull for Synchrony} \\ & + \text{Pull for Segregation} \\ & + \text{Order}|\text{Dyad}). \end{aligned} \quad (2)$$

Results indicate that our model accounted for 6.71% of the variance of task performance. There was a significant interaction between IBI synchrony and the pull for segregation ($F = 7.22, p = .02$). As predicted, higher IBI synchrony was associated with lower performance in high-segregation contexts but with higher performance in low-segregation contexts (Figure 4). Contrary to our hypothesis, we found no interaction effect of the pull toward synchrony in the context and IBI synchrony ($F = 2.12, p = .22$). There was also no three-way interaction ($F = .05, p = .82$). For full details of the model, see Supplemental Table S2 and Figure S2.

These results support the prediction that the function of IBI synchrony is context dependent. In high-segregation contexts, tightly coupled physiological rhythms may hinder effective coordination by reducing complementary task roles, whereas in low-segregation contexts, they may facilitate shared focus and coordination.

Figure 4

Performance as a Function of IBI Synchrony and the Pull for Segregation



Note. The y-axis represents performance, while the x-axis represents IBI synchrony. The different lines indicate conditions of high and low pull for segregation. IBI synchrony is associated with better performance under low segregation, but with worse performance under high segregation. IBI = interbeat interval. See the online article for the color version of this figure.

The Associations Between Cohesion and IBI Synchrony Depend on the Context's Requirement to Segregate and Synchronize

We next tested whether the relationship between IBI synchrony and participants' reported group cohesion varied across contextual pulls. An MLM model was used (Equation 3):

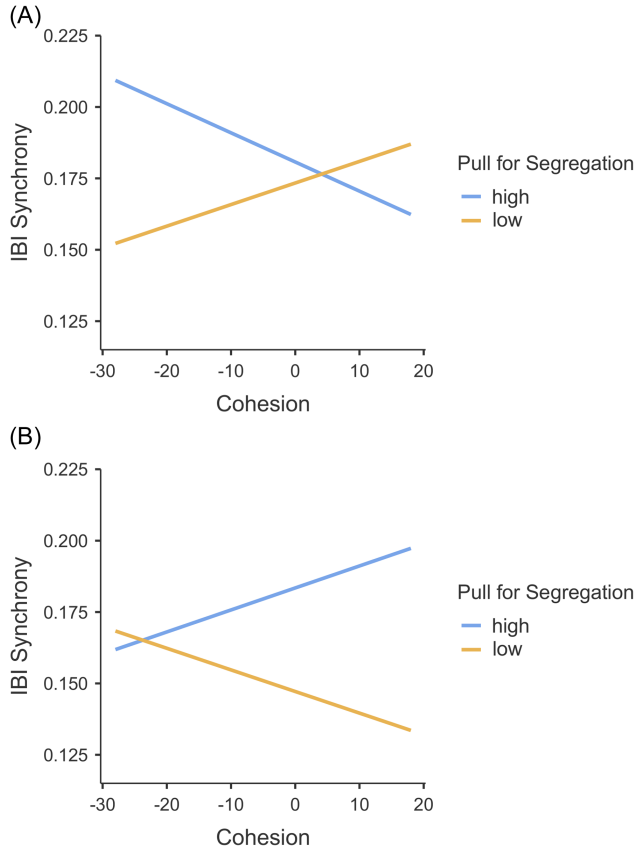
$$\begin{aligned} \text{IBI Synchrony} \sim & 1 + \text{Pull for Synchrony} \\ & + \text{Pull for Segregation} + \text{Order} \\ & + \text{Cohesion} + \text{Cohesion} \\ & \times \text{Pull for Synchrony} + \text{Cohesion} \\ & \times \text{Pull for Segregation} + \text{Cohesion} \\ & \times \text{Pull for Synchrony} \\ & \times \text{Pull for Segregation} \\ & + (1 + \text{Pull for Synchrony} \\ & + \text{Pull for Segregation} \\ & + \text{Order}|\text{Dyad}). \end{aligned} \quad (3)$$

Results indicate that our model accounted for 1.63% of the variance of IBI synchrony. As we hypothesized, we found a three-way interaction + Pull Toward Synchrony $\times$ Pull for Segregation $\times$ Cohesion ($F = 8.32, p = .01$). However, there was no significant interaction of Cohesion $\times$ Pull for Segregation and Cohesion $\times$ Pull Toward Synchrony, indicating that the association between cohesion and IBI synchrony depends on both pulls of the context (for full details of the model, see Supplemental Table S3 and Figure S3).

To interpret this interaction, we examined the slopes of cohesion at each level of the two pulls (Figure 5). In metastable contexts (high synchrony and high segregation or low synchrony and low segregation), cohesion was negatively associated with IBI synchrony. By contrast, in non-

Figure 5

IBI Synchrony as a Function of Cohesion, the Pull for Synchrony, and the Pull for Segregation



Note. The y-axis represents IBI synchrony, while the x-axis represents cohesion. The different lines indicate conditions of high and low pull for segregation. (A) The relationships under high-synchrony conditions; (B) the relationships under low-synchrony conditions. The patterns reveal that the relationship between IBI synchrony and cohesion depends on the contextual combination of synchrony and segregation. (A) Pull for synchrony = high. (B) Pull for synchrony = low. IBI = interbeat interval. See the online article for the color version of this figure.

meta-stable contexts (high synchrony and low segregation or low synchrony and high segregation), cohesion was positively associated with IBI synchrony.

These results suggest that IBI synchrony may support cohesion only in non-meta-stable contexts but be less aligned with perceived group bonding when demands are symmetrical, highlighting the context-dependent nature of its function.

The Associations Between Task Performance and EDA Synchrony Depend on the Context's Requirement to Synchronize and Segregate

To explore the role of sympathetic synchrony, we tested how EDA synchrony relates to task performance across contextual pulls. An MLM model was used (Equation 4):

$$\begin{aligned} \text{Task Performance} \sim & 1 + \text{Pull for Synchrony} \\ & + \text{Pull for Segregation} + \text{Order} \\ & + \text{EDA Synchrony} \\ & + \text{Pull for Synchrony} \\ & \times \text{Pull for Segregation} \\ & + \text{EDA Synchrony} \\ & \times \text{Pull for Synchrony} \\ & + \text{EDA Synchrony} \\ & \times \text{Pull for Segregation} \\ & + \text{EDA Synchrony} \\ & \times \text{Pull for Synchrony} \\ & \times \text{Pull for Segregation} \\ & + (1 + \text{Pull for Synchrony} \\ & + \text{Pull for Segregation} \\ & + \text{Order} | \text{Dyad}). \end{aligned} \quad (4)$$

Results indicate that our model accounted for 21.6% of the variance of task performance. There was an overall positive main effect for EDA synchrony ($F = 19.55, p < .001$). The interaction EDA Synchrony $\times$ Pull Toward Synchrony was significant ($F = 41.29, p < .001$), indicating that in contexts of high pull for synchrony, the association is negative, while in contexts of low pull for synchrony, the association is positive (Figure 6). The interaction between EDA synchrony and the pull for segregation was also significant ($F = 37.68, p < .001$), indicating that in contexts of high pull for segregation, the association is positive, while in contexts of low pull for segregation, the association is negative (Figure 7). The three-way interaction Pull Toward Synchrony $\times$ Pull for Segregation $\times$ EDA Synchrony was also significant ($F = 148.79, p < .001$). For full details of the model, see Supplemental Table S4 and Figures S4 and S5.

These findings suggest that EDA synchrony may be more helpful in contexts with low demands for behavioral coordination, presumably to compensate for the lack of behavioral alignment.

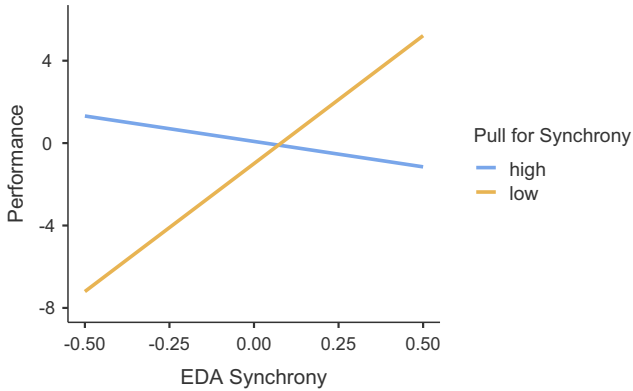
The Associations Between Cohesion and EDA Synchrony

Next, we tested whether EDA synchrony was associated with perceived cohesion, under varying contexts. An MLM model was used (Equation 5):

$$\begin{aligned} \text{EDA Synchrony} \sim & 1 + \text{Pull for Synchrony} \\ & + \text{Pull for Segregation} + \text{Order} \\ & + \text{Cohesion} + \text{Cohesion} \\ & \times \text{Pull for Synchrony} + \text{Cohesion} \\ & \times \text{Pull for Segregation} + \text{Cohesion} \\ & \times \text{Pull for Synchrony} \\ & \times \text{Pull for Segregation} \\ & + (1 + \text{Pull for Synchrony} \\ & + \text{Pull for Segregation} \\ & + \text{Order} | \text{Dyad}). \end{aligned} \quad (5)$$

Figure 6

Performance as a Function of EDA Synchrony and the Pull for Synchrony



Note. The y-axis represents performance, while the x-axis represents EDA synchrony. The different lines indicate conditions of high and low pull for synchrony. EDA synchrony predicts better performance under low-synchrony contexts, but not under high-synchrony contexts. EDA = electrodermal activity. See the online article for the color version of this figure.

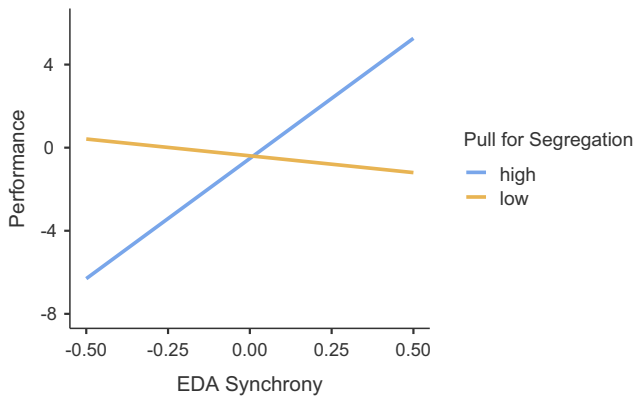
As hypothesized, we found no interaction effects of cohesion and no main effect of cohesion (for full details of the model, see Supplemental Table S5), indicating that cohesion and EDA synchrony are not associated under any context.

The Associations Between IBI Synchrony and EDA Synchrony Depend on the Context's Requirement

Finally, we tested how the association between IBI synchrony and EDA synchrony varied across contextual pulls. An MLM model was used (Equation 6):

Figure 7

Performance as a Function of EDA Synchrony and the Pull for Segregation



Note. The y-axis represents performance, while the x-axis represents EDA synchrony. The different lines indicate conditions of high and low pull for segregation. EDA synchrony predicts better performance under high-segregation contexts, but not under low segregation contexts. EDA = electrodermal activity. See the online article for the color version of this figure.

EDA Synchrony $\sim 1 + \text{Pull for Synchrony}$

+ Pull for Segregation + Order
 + IBI Synchrony + IBI Synchrony
 $\times$ Pull for Synchrony
 + IBI Synchrony
 $\times$ Pull for Segregation
 + IBI Synchrony
 $\times$ Pull for Synchrony
 $\times$ Pull for Segregation
 + (1 + Pull for Synchrony
 + Pull for Segregation
 + Order|Dyad).

(6)

Results indicate that our model accounted for 17.8% of the variance of EDA synchrony. There was a main effect of IBI synchrony ($F = 30.64, p < .001$), indicating an overall positive association between IBI synchrony and EDA synchrony. Moreover, there were three interactions: IBI Synchrony $\times$ Pull for Segregation ($F = 186.28, p < .001$), IBI Synchrony $\times$ Pull for Synchrony ($F = 84.67, p < .001$), and IBI Synchrony $\times$ Pull for Synchrony $\times$ Pull for Segregation ($F = 12.54, p < .001$). Figure 8 illustrates the association between IBI synchrony and EDA synchrony under each combination of pulls for synchrony and segregation. It seems that the association between IBI synchrony and EDA synchrony is more prominent in the more metastable contexts (high pull for synchrony and segregation/low pull for synchrony and segregation) and less in the non-meta-stable contexts. Additionally, when there is a high pull for synchrony or segregation, the association is positive, whereas in contexts wherein both are negative, the association is negative. Overall, while IBI synchrony and EDA synchrony are associated, their association also depends on the contextual pulls for synchrony and segregation. For full details of the model, see Supplemental Table S6 and Figure S7.

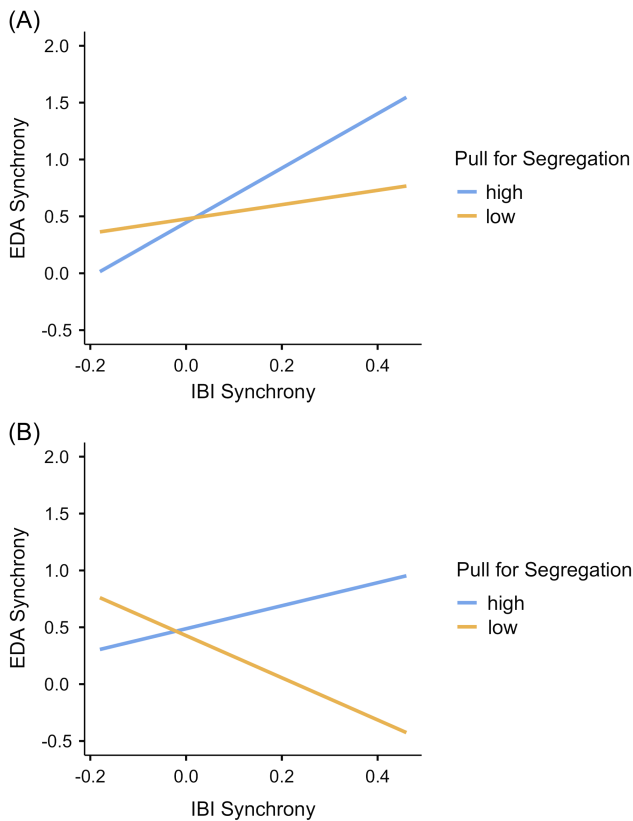
Additional Analyses

To test the robustness of our findings, we conducted an additional control analysis including dyadic demographic variables (gender composition, age, education, and religiosity) as fixed effects in the performance model. None of these variables were significant predictors (all $ps > .49$), and their inclusion did not affect the core results. This suggests that our main effects are not confounded by dyadic background factors (see Supplemental Table S7).

To ensure that condition-level differences in EDA synchrony were not confounded by baseline arousal, we conducted an exploratory analysis comparing mean EDA levels across conditions. This analysis, reported in the Supplemental Table S8, revealed no significant differences in baseline EDA.

To confirm that participants adopted a coordinated strategy during Training Phase 1, we conducted an exploratory analysis showing that behavioral synchrony, measured by Pearson

Figure 8
EDA Synchrony as a Function of IBI Synchrony, the Pull for Synchrony, and the Pull for Segregation



Note. The y-axis represents EDA synchrony, while the x-axis represents IBI synchrony. The different lines indicate conditions of high and low pull for segregation. (A) The relationships under high-synchrony conditions, (B) the relationships under low-synchrony conditions. The association between IBI synchrony and EDA synchrony depends on the contextual pulls for synchrony and segregation. (A) Pull for synchrony = high. (B) Pull for synchrony = low. EDA = electrodermal activity; IBI = interbeat interval. See the online article for the color version of this figure.

correlation of angular position, increased in the last 30 s in that phase compared with the rest of the time, $t(37) = 3.15$, $p < .01$ (see Supplemental Figure S8).

To further explore the joint role of physiological synchrony modalities, we tested whether EDA synchrony and IBI synchrony interact to predict performance. An exploratory MLM revealed a significant interaction between the two ($F = 7.94$, $p \leq .01$) and a significant three-way interaction with the pull for synchrony ($F = 7.12$, $p \leq .01$). These results suggest that the impact of one synchrony modality may depend on the level of the other and that this interaction is shaped by contextual demands. The full model and statistical details are provided in Supplemental Table S9.

To assess whether the contextual condition could be inferred from patterns of synchrony, we conducted an exploratory classification analysis using a decision tree model. Behavioral synchrony, IBI synchrony, and EDA synchrony were entered

as predictors of the four experimental conditions (High/Low Pull for Synchrony $\times$ High/Low Pull for Segregation). The model achieved 74.2% classification accuracy (95% CI: 69.3%–78.6%) with a Cohen's κ of .66, indicating substantial agreement beyond chance. These results suggest that multimodal synchrony reliably reflects the latent contextual structure, providing converging evidence for the experimental manipulation and theoretical framework.

To examine how synchrony evolved over time, we conducted an exploratory time-series analysis in which each 2-min task condition was divided into four 30-s bins. For each bin, we computed average behavioral, IBI, and EDA synchrony. The resulting time course patterns, displayed in Supplemental Figure S9, suggest that, when averaged across all dyads, synchrony appeared relatively stable over time in all modalities. However, this apparent stability reflects the aggregation across dyads; within individual dyads, fluctuations are to be expected and may be substantial. A detailed examination of these in-triad dynamics is beyond the scope of the present article, but this initial analysis provides a first step toward understanding the interplay between synchrony across modalities over time.

Discussion

The present study examined how the contextual pulls for synchrony and segregation shape the associations between physiological synchrony and social and performance outcomes. The findings offer initial empirical support for the proposal that these two latent pulls shape the function of interpersonal synchrony (Gordon et al., 2025; Mayo & Gordon, 2020). Specifically, the results suggest that physiological synchrony is not inherently beneficial or detrimental, but its impact on performance and cohesion depends on the broader contextual demands for behavioral alignment or differentiation.

A key contribution of this study is the development of an adapted experimental paradigm that enables the investigation of how contextual forces shape interpersonal synchrony. While prior studies examined interpersonal synchrony in various contexts (e.g., Danyluck & Page-Gould, 2019; Mayo et al., 2023; Paxton & Dale, 2013; Zhang et al., 2021), it has remained unclear what latent principles govern the effect of synchrony. To address this gap, we adapted a dyadic coordination task to independently manipulate the pull for synchrony and the pull for segregation. Our results confirmed that these manipulations successfully altered behavioral synchrony, demonstrating that these contextual forces have a tangible impact on coordination processes. Specifically, in contexts with a high pull for synchrony, there were higher levels of behavioral synchrony compared with contexts with a low pull for synchrony, while in contexts with a high pull for segregation, there were lower levels of behavioral synchrony compared with contexts with a low pull for segregation. This paradigm offers a more ecologically valid and flexible framework for studying synchrony in interactive

social environments, paving the way for future research on how individuals balance competing demands for alignment and autonomy in real-world interactions.

Beyond behavioral validation, we also tested whether multimodal synchrony could reliably capture the contextual structure of the task. Using behavioral synchrony, IBI synchrony, and EDA synchrony as predictors, the model successfully classified the experimental conditions, indicating substantial agreement beyond chance. These results suggest that multimodal synchrony reliably reflects the latent contextual structure, providing converging evidence for the experimental manipulation and theoretical framework.

Our findings also underscore the importance of considering multiple physiological measures when studying interpersonal physiological synchrony. IBI and EDA synchrony reflect distinct autonomic processes (Mayo et al., 2021; Palumbo et al., 2017), with IBI governed by both the sympathetic nervous system and the parasympathetic nervous system (Shaffer & Ginsberg, 2017) and associated with social bonding and coregulation (Feldman, 2012; Feldman et al., 2011; Gordon et al., 2020; Mayo et al., 2021), while EDA is a marker of the sympathetic nervous system activation (Dawson et al., 2007), linked to shared arousal and attentional engagement (Critchley, 2002).

IBI Synchrony and Performance

As hypothesized, the relationship between IBI synchrony and performance was moderated by the pull for segregation. In high-segregation contexts, IBI synchrony was negatively associated with performance, whereas in low-segregation contexts, it was positively associated with performance. This aligns with the idea that IBI synchrony is primarily linked to a shared emotional state and a sense of being on the same page (Feldman, 2012; Feldman et al., 2011), which can enhance performance when individuals are expected to work closely together. However, when autonomy and independent contributions are required, heightened IBI synchrony may be detrimental, possibly by marking a lower experience of differentiation between dyad members, when such a differentiation is actually called for (Gordon et al., 2025; Mayo & Gordon, 2020).

Interestingly, the pull for synchrony did not moderate the IBI-performance relationship, contrary to our predictions. This suggests that segregation demands exert a stronger influence on IBI synchrony than explicit demands for synchrony in the current task. One possibility is that the current task pull for segregation constrains physiological coregulation more than shared goals or behavioral alignment does. That is, it might be possible that the manipulation of segregation in the context of the task was stronger than the manipulation of the pull for synchrony. Future research should refine experimental designs to further disentangle

the relative influences of these two latent contextual potentials.

IBI Synchrony and Cohesion

A more complex pattern emerged for the relationship between IBI synchrony and cohesion. As predicted, in high-synchrony contexts, the pattern of results mirrored the IBI-performance findings: IBI synchrony was positively associated with cohesion when segregation was low but negatively associated when segregation was high. However, in low-synchrony contexts, the opposite pattern emerged: IBI synchrony was positively associated with cohesion in high-segregation contexts and negatively associated in low-segregation contexts.

These results suggest that IBI synchrony aligns with cohesion only in non-meta-stable contexts, meaning when there is minimal tension between the pulls for synchrony and segregation (Gordon et al., 2025; Mayo & Gordon, 2020). By contrast, in metastable contexts—where both pulls are competing—individuals may desynchronize to navigate the conflicting demands of the situation. Notably, cohesion was measured after task completion, meaning that dyads that reported higher cohesion were those who exhibited IBI synchrony in non-meta-stable contexts but desynchronized in metastable contexts. This suggests that previous findings linking IBI synchrony with social bonding may be context specific—emerging mainly in high-synchrony, low-segregation settings (e.g., parent–child relationships; Feldman, 2012; Feldman et al., 2011). Our result also highlights that these two fundamental pulls are not two opposites of one dimension but two latent potentials that can work independently to exert results, which is in line with the model of multimodal flexible synchrony (Gordon et al., 2025; Mayo & Gordon, 2020).

EDA Synchrony and Performance

As hypothesized, the association between EDA synchrony and performance was moderated by the contextual pulls for synchrony and segregation. Consistent with the flexible synchrony framework (Gordon et al., 2025; Mayo & Gordon, 2020), EDA synchrony was positively associated with performance in high-segregation or low-synchrony contexts and negatively associated with performance in low-segregation or high-synchrony contexts (see also Dale et al., 2020).

These results suggest that EDA synchrony may serve a compensatory function in contexts wherein overt behavioral coordination is limited. When individuals must act independently (high segregation) or where alignment is discouraged (low synchrony), shared physiological arousal may maintain attentional focus and mutual engagement, thus supporting performance. Conversely, when coordination is already high, excessive arousal may indicate overstimulation or reduced flexibility, potentially impairing effective role differentiation. Notably, the strongest positive association emerged in

high-segregation, low-synchrony contexts, highlighting how physiological alignment can support shared cognitive engagement even in the absence of behavioral synchrony. Importantly, exploratory analyses revealed no significant differences in mean EDA levels across the four experimental conditions, suggesting that the observed effects of EDA synchrony on performance are unlikely to be driven by overall differences in arousal across contexts.

EDA Synchrony and Cohesion

Unlike IBI synchrony, EDA synchrony was not significantly associated with cohesion. Previous results regarding the association between EDA synchrony and relationship outcomes were inconsistent (e.g., Behrens et al., 2020; Chikersal et al., 2017; Gordon, Wallot, & Berson, 2021; Mønster et al., 2016). The current findings suggest that EDA synchrony is more closely related to shared task engagement, immediate emotional exchanges, and attentional alignment rather than long-term relational bonding. The absence of association with cohesion suggests that sympathetic arousal alone is not a sufficient predictor of social connectedness. Future research should explore whether EDA synchrony's role in cohesion depends on task structure, emotional intensity, or social context.

Exploratory Analysis: Association Between IBI and EDA Synchrony

An exploratory analysis revealed a complex interaction between IBI and EDA synchrony across different contexts. Overall, there was a positive main effect, indicating that, across conditions, IBI and EDA synchrony were generally positively associated. However, this relationship varied depending on contextual pulls for synchrony and segregation.

In all contexts, the association between IBI and EDA synchrony was positive, except for the context with low pulls for synchrony and segregation. In this context, the association was negative, potentially indicating that there were low demands for effort to coordinate or to act independently. Thus, EDA synchrony might reflect unnecessary mutual arousal that goes hand in hand with lower social bonding, as reflected in IBI synchrony.

Limitations and Future Directions

Notwithstanding its contributions, this study has several limitations. First, cohesion was measured only once, after task completion, making it difficult to assess how cohesion evolved throughout the task. Future research should incorporate multiple time points to capture dynamic changes in perceived cohesion. Second, the study was conducted in a controlled laboratory environment, which, while allowing for precise manipulation of synchrony and segregation, may not fully generalize to real-world social interactions. Future

studies could examine how these processes unfold in naturalistic team settings or clinical contexts. Third, our sample consists mainly of young women, which makes it difficult to generalize to other populations. Fourth, another limitation concerns the power analysis approach. While this sample size may not provide full power to detect very small effects, the robust and theoretically coherent effects observed across models suggest that the study was sufficiently powered to detect the key patterns of interest. We acknowledge that this method does not fully capture the complexity of our multilevel modeling framework with dyads and repeated measures. Simulation-based power analysis would provide more precise estimates but requires assumptions about variance components and intraclass correlations that are difficult to estimate reliably in novel, interactive designs. Despite this, the fact that we observed robust and theoretically meaningful effects suggests that the sample size was sufficient to detect key patterns. Future research with larger samples and simulation-based power planning could further strengthen the empirical foundation of this framework. Relatedly, due to the model's complexity and inclusion of random slopes, we were unable to report effect sizes for individual predictors (e.g., partial R^2). We instead reported marginal R^2 values as an index of model explanatory power. Fifth, the present study does not include subjective task difficulty ratings, which could help disentangle whether physiological synchrony reflects shared cognitive engagement, stress, or task demand. Future studies could benefit from collecting such ratings to better contextualize physiological data. Sixth, while our present analyses focused on task-aggregated synchrony levels, future work could benefit from examining the temporal dynamics of synchrony across the interaction. Recent studies suggest that a balance between synchrony and novelty over time predicts more adaptive outcomes (Ravreby et al., 2022). Incorporating such time-sensitive metrics may allow us to test how the flexible synchrony framework operates in real-time regulatory processes, including the emergence, breakdown, and reestablishment of alignment under varying contextual pulls. Finally, the study focused on dyadic interactions, raising the question of whether similar dynamics apply to larger groups. Future research should explore whether the balance of synchrony and segregation influences group-level coordination in more complex social networks.

Conclusion

The present study offers preliminary empirical support for the multimodal flexible synchrony (Gordon et al., 2025; Mayo et al., 2021), which postulates that social context can be characterized by their pulls of synchrony and segregation and that these pulls affect the meaning and outcomes of interpersonal synchrony. Specifically, it offers a novel framework for understanding the dynamic role of physiological synchrony in social interactions by incorporating

latent contextual forces of synchrony and segregation. Our findings reveal that the IBI and EDA synchrony serve different functions and that their association with cohesion and performance depends on the contextual demand for synchrony and segregation. These findings pave the way for future research to further explore how physiological synchrony interacts with situational demands across diverse social settings.

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